Physics-Based Validation Workflow

Predicted Flow Field
(U – Net Output)

Gate 1: Incompressibility

$\nabla \cdot u \leq tolerance?$

Pass

Fail (output rejected

Gate 2: Energy Spectrum

E(k) within ±10% of DNS?

Pass

Fail (output rejected

Gate 3: Wall Law

U+ ~ log(y+), κ,B in range?

Pass

Fail (output rejected

All Physic Gates Passed (Admissible Prediction)

Physics Checks:

- Mass Conservation

  ∇ · u = 0",
- Energy Spectrum

  Kolmogorov -5/3 law
- Wall Law

  U+ = 1/κ ln(y+) + B"

Validation Benefits:

- Enforces physical consistency
- Rejects non-physical predictions
- Improves reliability of ML models

Note: *Band-wise accuracy checking can be applied\nto flag/suppress predictions in specific y+ regions*